

MCT 211

Automatic Control

Lecture 2: Process Dynamics

Mechatronics Dep.
Faculty of Engineering
Ain Shams University

By Dr. Diaa Emad

Process Dynamics

Diesel Engine

The equation of motion of the engine may be written as:

$$J_n \frac{d\omega}{dt} = T_D - T_R$$

J_n = the moment of inertia of all rotating masses

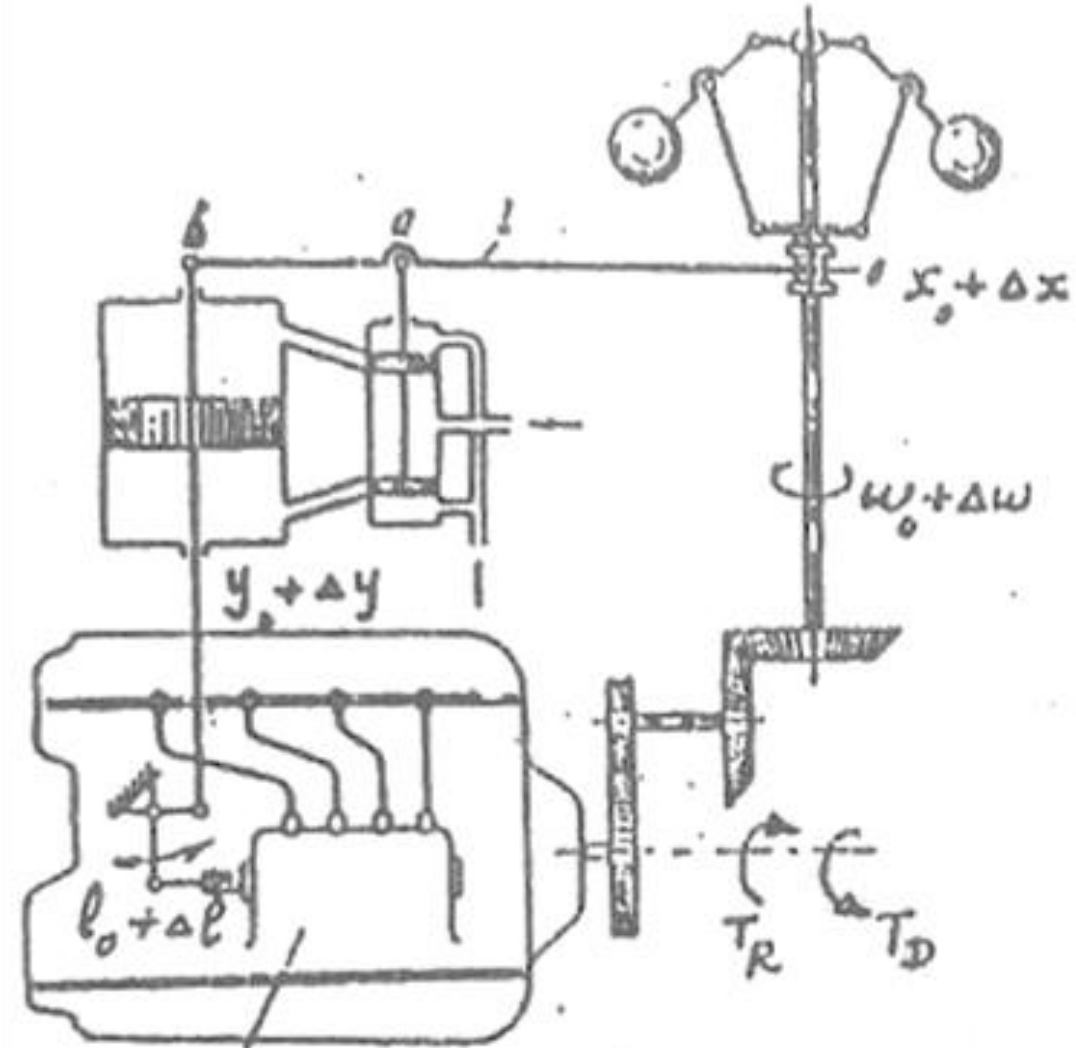
ω = the angular velocity of the engine

T_D = the engine driving torque, $T_D = T_D(\omega, l)$

T_R = the resisting torque, $T_R = T_R(\omega)$

l = the position of the fuel rack.

$$J_n \frac{d\omega}{dt} = T_D(\omega, l) - T_R(\omega)$$



Process Dynamics

Diesel Engine

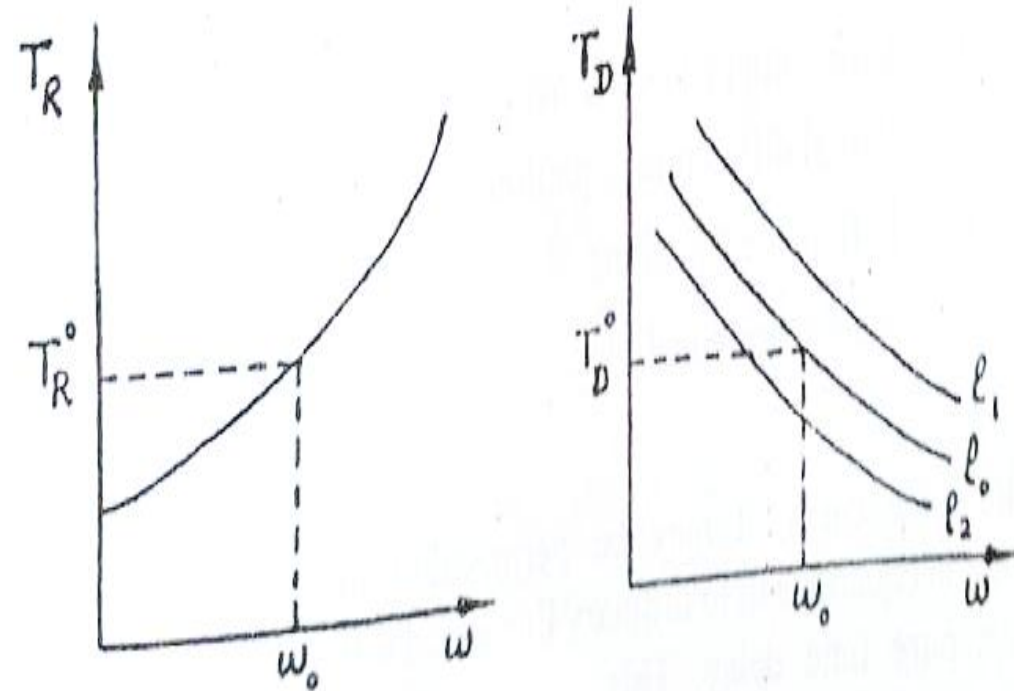
- The relations of driving and resisting torques can be determined experimentally or analytically as shown in Figure.
- Generally, the characteristics of $T_D(\omega, l)$ and $T_R(\omega)$ are non-linear, from which it follows that the mathematical model is **non-linear differential equation**.
- If we assume that all torques and also the angular velocity can take small variations about some steady state values, then

$$T_D = T_D^0 + \Delta T_D$$

$$T_R = T_R^0 + \Delta T_R$$

$$\omega = \omega^0 + \Delta\omega$$

$$J_n \frac{d\omega}{dt} = T_D(\omega, l) - T_R(\omega)$$



Process Dynamics

Diesel Engine

- Assuming also that the engine has monotonic characteristics around the **normal operating conditions**, then these characteristics may be linearized by using Taylor's expansion around the normal operating point.
- Neglecting the terms of the second and higher orders, then

$$T_D = T_D^o + \left. \frac{\partial T_D}{\partial \omega} \right| \Delta \omega + \left. \frac{\partial T_D}{\partial l} \right| \Delta l$$

$$T_R = T_R^o + \left. \frac{\partial T_R}{\partial \omega} \right| \Delta \omega$$

- Considering $T_D^o = T_R^o$ at steady state, then:

$$J_n \frac{d(\Delta \omega)}{dt} + \left(\left. \frac{\partial T_R}{\partial \omega} \right| - \left. \frac{\partial T_D}{\partial \omega} \right| \right) \Delta \omega = \left. \frac{\partial T_D}{\partial l} \right| \Delta l$$

Process Dynamics

Diesel Engine

- It is easy to formulate the last expression in the form:

$$T \frac{d(\Delta\omega)}{dt} + \Delta\omega = K\Delta l$$

- T is the time constant of the engine.
- K is the gain of the engine.

Process Dynamics

Hydraulic Tank

The balance equation of the hydraulic tank may be written as:

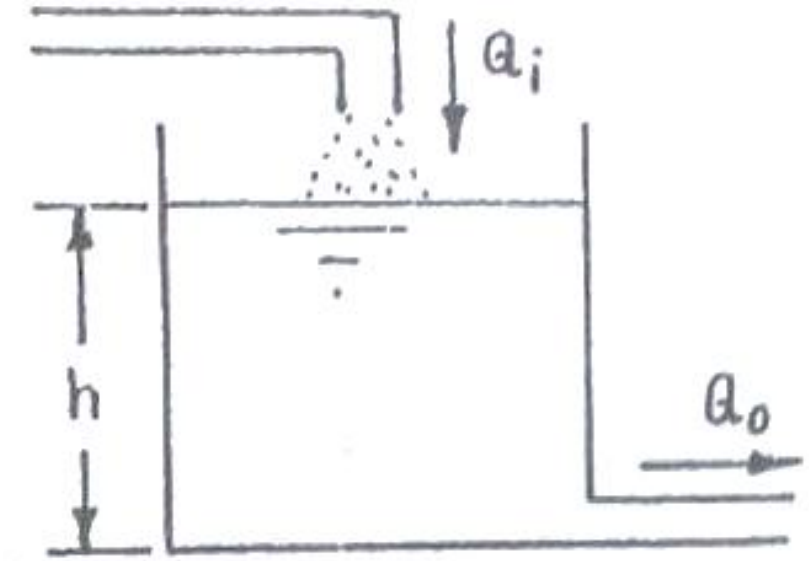
$$A \frac{dh}{dt} = Q_i - Q_o$$

A = the cross-sectional area of the tank

h = the head

Q_i = the input flow rate

Q_o = the output flow rate



Process Dynamics

Hydraulic Tank

- Considering small variations of Q_i , Q_o , h about the steady state operating conditions, then

$$Q_i = Q_i^o + \Delta Q_i$$

$$Q_o = Q_o^o + \Delta Q_o$$

$$h = h^o + \Delta h$$

- But the output flow rate Q_o can be expressed in terms of the head h as follows:

$$Q_o = Q_o(h) = k\sqrt{h}$$

$$Q_o = Q_o^o + \left. \frac{\partial Q_o}{\partial h} \right| \Delta h$$

Process Dynamics

Hydraulic Tank

- Considering the last relations, the mathematical model can be written in the form:

$$A \frac{dh}{dt} + \frac{\partial Q_o}{\partial h} \Delta h = \Delta Q_i$$

- It is easy to formulate the last expression in the form:

$$T \frac{dh}{dt} + \Delta h = K \Delta Q_i$$

- T is the time constant of the tank.
- K is the gain of the tank.

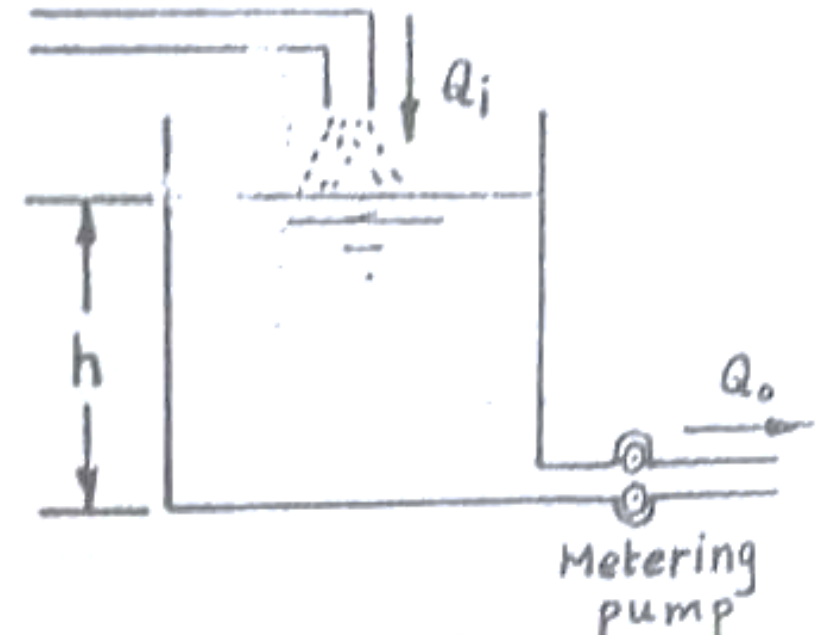
Process Dynamics

Hydraulic Tank with metering pump

- The derivation of the mathematical model in this case is similar to the previous one, but the output flow rate Q_o is constant and independent on the head h .
- Then it is easy to get the head in this case at any time instant by the relation:

$$h = \frac{1}{A} \int (Q_i - Q_o) dt$$

This expression means that any difference between Q_i and Q_o will lead to continuous increase or decrease in the head h .



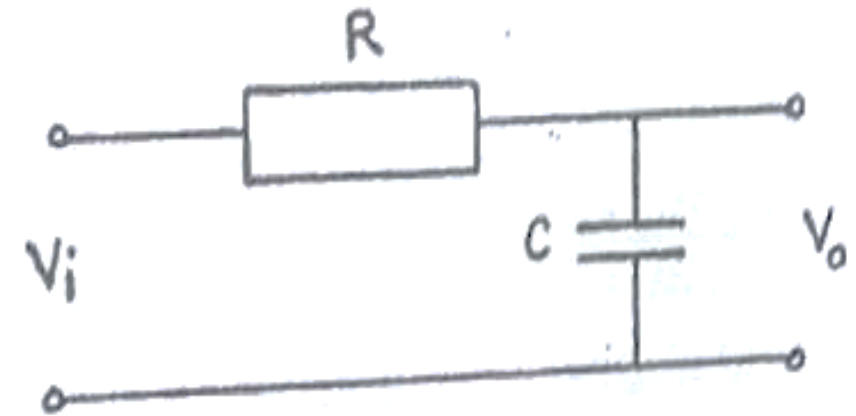
Process Dynamics

RC Circuit

- It is easy to prove that the relation between the input and output voltages of the RC circuit in the form:

$$T \frac{dV_o}{dt} + V_o = V_i$$

- T is the RC time constant.
- V_i is the input voltage.
- V_o is the output voltage.



Process Dynamics

Regulation of Plant

- The plants which can be described by a first order linear or linearized differential equations; as given in the previous examples; are termed “Single capacity plants”.
- The mathematical models of diesel engine, hydraulic tank and RC circuit are self-regulating plants, while the mathematical model of hydraulic tank with metering pump is non-self-regulating plant.

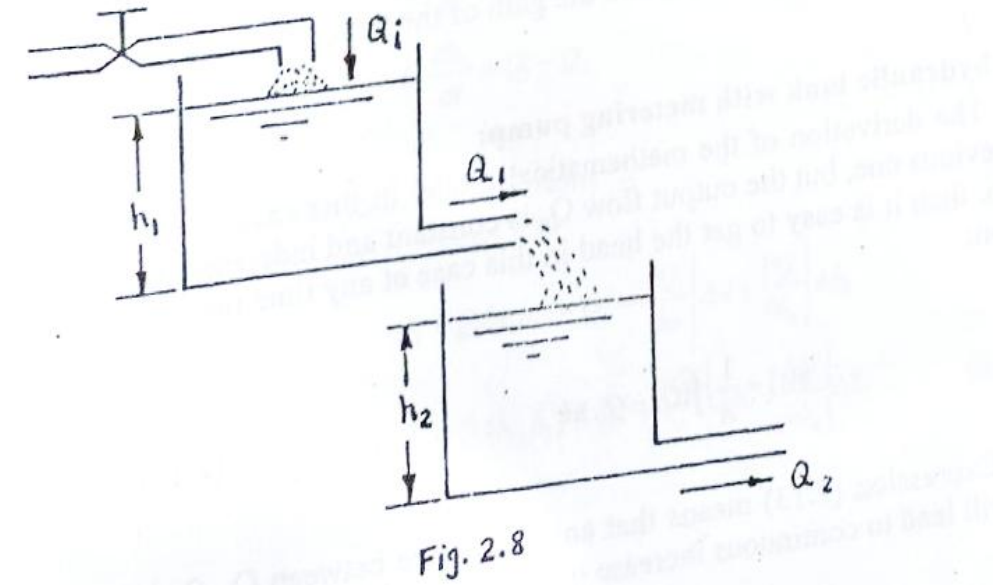
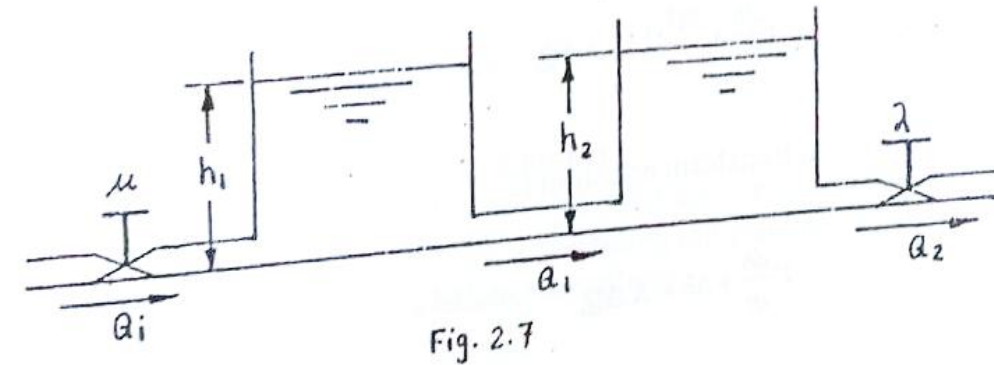
Process Dynamics

Two Capacity Plants: Two Hydraulic Tanks

- The system of two tanks may be connected either as shown in Fig. (2.7) or in Fig. (2.8).
- Considering the plant in Fig. (2.7), it is easy to get the following mass balance equations:

$$A_1 \frac{dh_1}{dt} = Q_i - Q_1$$

$$A_2 \frac{dh_2}{dt} = Q_1 - Q_2$$



Process Dynamics

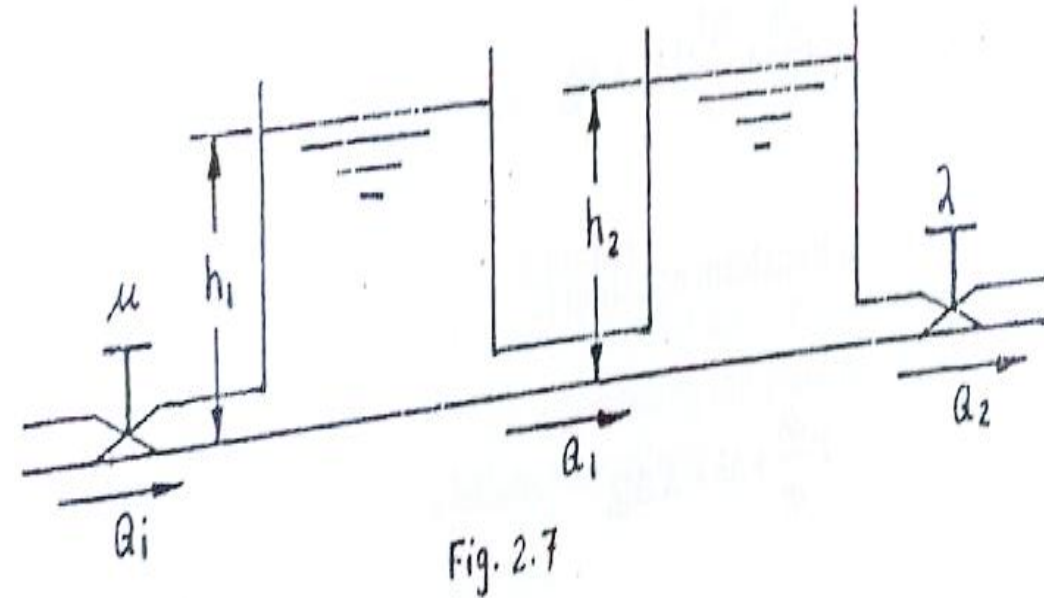
Two Capacity Plants: Two Hydraulic Tanks

- We have the following relations:

$$Q_i = Q_i(\mu, h_1) = Q_i^o + \left. \frac{\partial Q_i}{\partial \mu} \right| \Delta \mu + \left. \frac{\partial Q_i}{\partial h_1} \right| \Delta h_1$$

$$Q_1 = Q_1(h_1, h_2) = Q_1^o + \left. \frac{\partial Q_1}{\partial h_1} \right| \Delta h_1 + \left. \frac{\partial Q_1}{\partial h_2} \right| \Delta h_2$$

$$Q_2 = Q_2(h_2, \lambda) = Q_2^o + \left. \frac{\partial Q_2}{\partial h_2} \right| \Delta h_2 + \left. \frac{\partial Q_2}{\partial \lambda} \right| \Delta \lambda$$



Process Dynamics

Two Capacity Plants: Two Hydraulic Tanks

- After substitution:

$$A_1 \frac{dh_1}{dt} = \left. \frac{\partial Q_i}{\partial \mu} \right| \Delta \mu + \left(\left. \frac{\partial Q_i}{\partial h_1} \right| - \left. \frac{\partial Q_1}{\partial h_1} \right| \right) \Delta h_1 - \left. \frac{\partial Q_1}{\partial h_2} \right| \Delta h_2$$

$$A_2 \frac{dh_2}{dt} = \left. \frac{\partial Q_1}{\partial h_1} \right| \Delta h_1 + \left(\left. \frac{\partial Q_1}{\partial h_2} \right| - \left. \frac{\partial Q_2}{\partial h_2} \right| \right) \Delta h_2 - \left. \frac{\partial Q_2}{\partial \lambda} \right| \Delta \lambda$$

- If the control engineer is interested in one of the variables either h_1 or h_2 , the given equations can be solved for any one of the variables.
- The solution of the two equations w.r.t. Δh_1 or Δh_2 will lead to a second order differential equation.

Process Dynamics

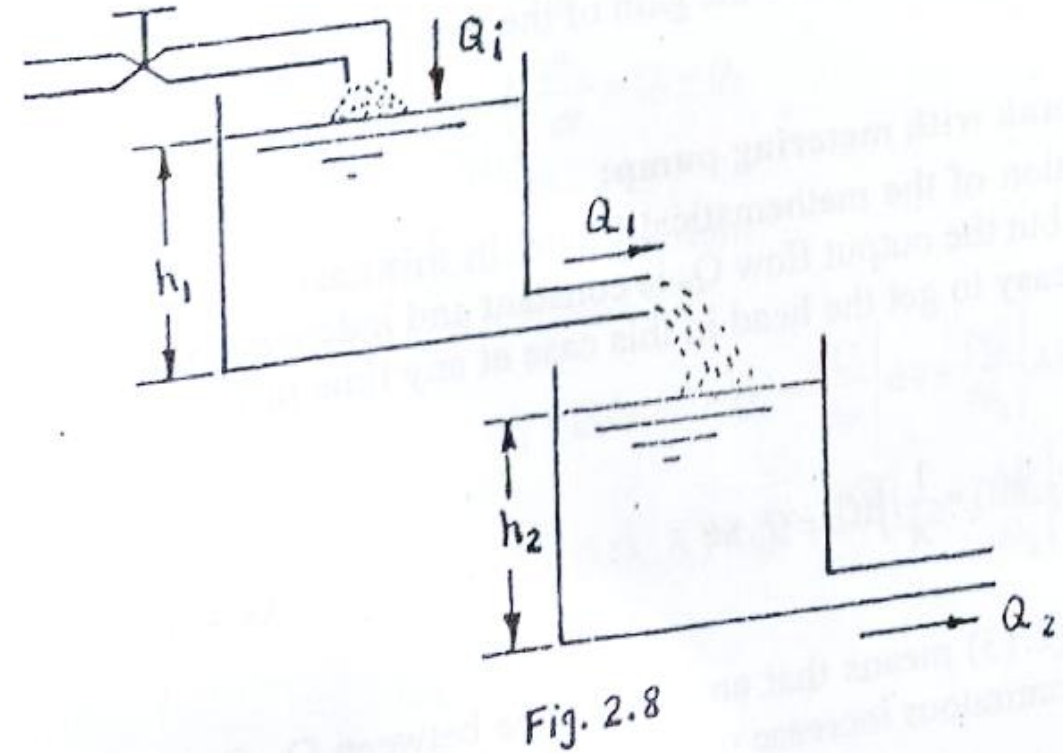
Two Capacity Plants: Two Hydraulic Tanks

- Considering the plant in Fig. (2.8):

$$Q_i = Q_i(\mu) = Q_i^o(\mu) + \left. \frac{\partial Q_i}{\partial \mu} \right| \Delta \mu$$

$$Q_1 = Q_1(h_1) = Q_1^o + \left. \frac{\partial Q_1}{\partial h_1} \right| \Delta h_1$$

$$Q_2 = Q_2(h_2) = Q_2^o + \left. \frac{\partial Q_2}{\partial h_2} \right| \Delta h_2$$



Process Dynamics

Two Capacity Plants: Two Hydraulic Tanks

- After substitution:

$$A_1 \frac{dh_1}{dt} = \left. \frac{\partial Q_i}{\partial \mu} \right| \Delta \mu - \left. \frac{\partial Q_1}{\partial h_1} \right| \Delta h_1$$

$$A_2 \frac{dh_2}{dt} = \left. \frac{\partial Q_1}{\partial h_1} \right| \Delta h_1 - \left. \frac{\partial Q_2}{\partial h_2} \right| \Delta h_2$$

- The set of equations can be solved either for Δh_1 or Δh_2 and it will result a second order differential equation.
- Comparing the two sets of equations for the two plants, it is concluded that in case of the first plant, the head h_1 affects the head h_2 and the head h_2 affects h_1 .
- But in case of the second plant, the head h_1 affects the head h_2 , while the head h_2 does not affects the head h_1 .
- For that reason, the plants similar to that in Fig. (2.7) are called “interacting plants” and the other type is called “non-interacting plants”.

Process Dynamics

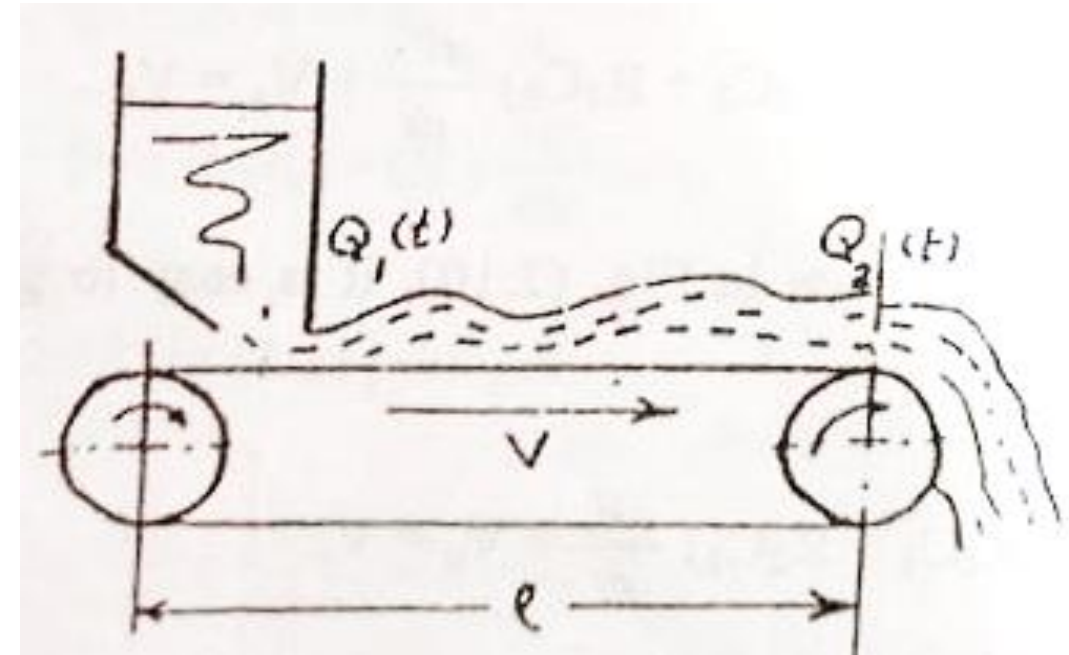
Pure Time Delay in Plants

- The piping connections included either in the plants or in the connected instruments usually leads to the appearance of a pure time delay in the system.
- Figure shows a belt conveyer, which can be used for transmitting solid material.
- The equation of mass balance in this case may be written in the form:

$$Q_2(t) = Q_1(t - \tau)$$

- τ is the pure time delay of the plant.

$$\tau = \frac{L}{V}$$



Process Dynamics

Table 1: System classifications according to their transfer functions

Transfer function	Classification
$W(S) = \frac{O(S)}{I(S)} = k$	Proportional or gain
$W(S) = \frac{O(S)}{I(S)} = \frac{k}{S}$	Integrator
$W(S) = \frac{O(S)}{I(S)} = kS$	Differentiator
$W(S) = \frac{O(S)}{I(S)} = e^{-s\tau}$	Pure time delay
$W(S) = \frac{O(S)}{I(S)} = \frac{k}{TS + 1}$	First order
$W(S) = \frac{O(S)}{I(S)} = \frac{TkS}{TS + 1}$	Real differentiator
$W(S) = \frac{O(S)}{I(S)} = k \left(\frac{aS + 1}{bS + 1} \right)$	Lead compensator if $a > b$ Lag compensator if $a < b$
$W(S) = \frac{O(S)}{I(S)} = \frac{\omega_n^2}{S^2 + 2\zeta\omega_n + \omega_n^2}$	Oscillatory 2 nd order (underdamping) if $\zeta < 1$ Aperiodic 2 nd order (overdamping) if $\zeta > 1$

Thank You

Any Questions ??